

TASK-12

9-7-26

Topic : Implementation of Bastion Host in AWS VPC (with Internet Gateway and NAT Gateway)

Step 1: Create a VPC

1. Go to VPC.
2. Click Create VPC.
3. Name: my-vpc
4. IPv4 CIDR: 10.0.0.0/16
5. Click Create VPC

☒	kittu vpc	vpc-095b4678357e6039e	Available	-	-	Off	10.0.0.0/16
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Step 2: Create Subnets

Public Subnet:

1. Name: public-subnet
2. VPC: my-vpc
3. CIDR: 10.0.1.0/24

Subnet 1 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv6 subnet CIDR block

Private Subnet:

1. Name: private-subnet
2. VPC: my-vpc
3. CIDR: 10.0.2.0/24

Subnet 2 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block
 256 IPs

You have successfully created 2 subnets: subnet-0a3258a55d156cd70, subnet-033bc3fe471d77430

Subnets (2) [Info](#) Last updated less than a minute ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

Subnet ID : subnet-0a3258a55d156cd70 Subnet ID : subnet-033bc3fe471d77430 Clear filters

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	privatesubnet1	subnet-033bc3fe471d77430	Available	vpc-095b4678357e6039e kitt...	Off	10.0.1.0/24
☐	publicsubnet1	subnet-0a3258a55d156cd70	Available	vpc-095b4678357e6039e kitt...	Off	10.0.0.0/24

Step 3: Create an Internet Gateway

1. Go to **Internet Gateways**
2. Click **Create Internet Gateway**
3. Name: my-igw
4. Select the Internet Gateway
5. Click **Attach to VPC**
6. Choose my-vpc

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

You can add 49 more tags.

🟢 The following internet gateway was created: igw-0af268cf3fc4c6078 - kittu igw. You can now attach to a VPC to enable the VPC to communicate with the internet.

igw-0af268cf3fc4c6078 / kittu igw

Step 4: Create a Public Route Table

1. Go to Route Tables
2. Create a Route Table
3. Name: public-rt
4. Select my-vpc

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

You can add 49 more tags.

Enter the Destination target internet gateway and save it

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	Internet Gateway	-	No	CreateRoute
	Q igw-0af268cf3fc4c6078			

Routes (2)					
Filter routes Both Edit routes					
Destination	Target	Status	Propagated	Route Origin	
0.0.0.0/0	igw-0af268cf3fc4c6078	Active	No	Create Route	
10.0.0.0/16	local	Active	No	Create Route Table	

Go to the subnet association edit it and save it

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Explicit subnet associations (0)

Edit subnet associations

Available subnets (1/2)

< 1 >

⚙

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	privatesubnet1	subnet-033bc3fe471d77430	10.0.1.0/24	-	Main (rtb-022fda8f661f2eec4)
☒	publicsubnet1	subnet-0a3258a55d156cd70	10.0.0.0/24	-	Main (rtb-022fda8f661f2eec4)

Selected subnets

subnet-0a3258a55d156cd70 / publicsubnet1

Cancel Save associations

You have successfully updated subnet associations for rtb-06d8cda18f8023ca8 / public routetable.

rtb-06d8cda18f8023ca8 / public routetable

Actions

Step 6: Create a Private Route Table

1. Create another Route Table.
2. Name: private-rt

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

private routetable

VPC

The VPC to use for this route table.

vpc-095b4678357e6039e (kittu vpc)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q private routetable

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

Routes (2)

Both [Edit routes](#)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-013cdbb60f8b4e0ff	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Go to the subnet association, edit it, and save it.

Edit subnet associations
Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	privatesubnet1	subnet-033bc3fe471d77430	10.0.1.0/24	-	Main (rtb-022fda8f661f2eec4)
☐	publicsubnet1	subnet-0a3258a55d156cd70	10.0.0.0/24	-	rtb-06d8cda18f8023ca8 / public route...

Selected subnets

subnet-033bc3fe471d77430 / privatesubnet1

Cancel Save associations

✓ You have successfully updated subnet associations for rtb-0d7a2e83e3b35995b / private routetable.

After the complete this, open the terminal

```
PS C:\Users\Karthik Reddy> cd downloads
PS C:\Users\Karthik Reddy\downloads> ssh -i "kittu.pem" ubuntu@18.207.135.27
The authenticity of host '18.207.135.27 (18.207.135.27)' can't be established.
ED25519 key fingerprint is SHA256:uX62qxrA0ohvtpKA0xj3RhTEARN5VrUN+cG8RlIIShc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? YES
Warning: Permanently added '18.207.135.27' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)
```

Create the file:

Command : touch file name

vi file name

```
ubuntu@ip-10-0-0-95:~$ sudo touch vishi.pem
ubuntu@ip-10-0-0-95:~$ vi vishi.pem
```

Open new tab in the terminal

Command for see the key: **cat file name.pem**

```
PS C:\Users\Karthik Reddy> cd downloads
PS C:\Users\Karthik Reddy\downloads> cat kittu.pem
-----BEGIN RSA PRIVATE KEY-----
MIIeowIBAAKCAQEAg/D/oHPmKeLNvVreTsBognJdF5dW0VhwcPHTkyRksTeaXx3f
yErtDLrWk9ISS/J+JRtki4KF37w900E6nYxv/67JFyb26SofHCd52gtNiVeRMnb3
kAA2yinRQd79SQt+qgCEKGptQBhU2IY+7LdMmK5hyMRfEhxbkwSnIKNRGbflNQOD
jpWCjSj+QfrsGiURrNdAY1a00ZKpEtQ8sek5WxX14a92TNnvcidhZMbfcJPFuU5e
36dwixQBUFraDZ/tavarfI2FJ+qfPW/HL0LNW8YA04qsgBMUX1FZjyNCZtT8/KPI
Xz4sLkcAh4iwsDE2iPMkLU3i1RgN402rpdYzlwIDAQABAoIBAHWElaMcNA8nR1T0
c4dN20B3vArYw/UT7l4z2tDx+B8iCt/B8mbaVyTh0rHz3c+M4Q0JqPH1dvzRkuKi
hkaSayZNV2H8nrwErxi2HoGaU9y6PiJIz7Hbg5cvd7fW6jUWST12dt2MyDkLqEoJ
j06Jhc/BOFJZbUVannhgsL4NAexZ9oI3G5iTJJj+rbPMLZhHJlXrAocZ7bpQwUpS
FyZEQAJbbly4FTmETzpJxuU8Tf2Jtt+7SZoWzNUS/vSZFmiXJ0h245Jg8gxouiaY
23PrJDPildGMF7NmOaFE1aWVLVGi3BibgkCwaKNzzyHzlLTNNFVc0fuA9N8glMhH
0xWJ1pkCgYEA0o2PezK0rXDtgCTohIXRgvqYlXxiKHx1W4QiRV0+T5Ki93ApA6/f
ORbUKtCVY1jLZUYEGr5+njbmq8kWoP+AT2ztYt/FB7yZiPKuWeTMFfomRldCN+N
1g4OYM4qXHeDCmkLpEnJRRWwhTtRRzRvQKH6j3wMzoqreZjyXCs6H0bMcgYEA0Q3k
8VtsKuN8HDRAQs2SMqT0c1mZ0t7MftVPS+4ziHENLY78A0gCyJuFZqhiRh/+Iqwk
zpPcXUhdZTYZSDvSpqGHubyFAB+6jwxesXgF/OyZfbgZb0GLyviKz91xp52DE6cm
RVhTgFf455Hs0wmj3VCsjpu+VkvRe2DxCH7J5o0CgYEANesoRYldumfUYys1EMm
zY3E6d3s9Lzh15k30jp6NyZBVVbshn2VJ960j2wvTAElaGpelli+0naNHquDxhV6
ugvveXX/ZMqkwKc61R+tiH6B+sHlBr4Y1vJDjaMl9RllUPSYgKgiUN8WT2mLweFN
D00+3tpwtWTZJILP7lHoc3cCgYAUoP1df7r9xHLjyZgpv2VU0Ag7NbU0JPYi8e0g
R4IYTaojCb/+Suazav1P8v+3kcKfI00Civ6syc5FhYAbK6Q0w1PEVUD+kSL0qLDJ
UjaNdb/bswX3f4102xW/i0eC1b0wfEb/gsbCnGu4ViL7dDESTRqoW9/9XVu8BAXk
47YCpQKBgALLPkh1eDTTKy6qhSoG6EaQL+pR3lvu4916LehbgHvaUYftQnA9zCFV
2Gb9XlhQoJshucOL8Rql+3F4Qwa98LNYvRquZaHpw8KezYnaZKpjHeE00ZwE2wrn
RReKdczKZaLRsNgOBpHlmFqOV6H0MnUEhmfp09dps9ShGVrLqIi8
-----END RSA PRIVATE KEY-----
PS C:\Users\Karthik Reddy\downloads> |
```

Copy the private key and paste it the back tab

```
j06Jhc/BOFJZbUVannhgsL4NAexZ9oI3G5iTJJj+rbPMlZhHJlXrAocZ7bpQwUpS
FyZEqAJbbly4FTmETzpJxuU8Tf2Jtt+7SZoWzNUS/vSZFmiXJ0h245Jg8gxouiaY
23PrJDPiIdGMF7Nm0aFE1aWVLVGi3BibgkCwaKNzzyHzlLTNNFVc0fuA9N8gLMhH
0xWJ1pkCgYEA0o2PezK0rXDtgCTohIXRgvqYlXxiKHx1W4QiRV0+T5Ki93ApA6/-
ORbUKtCVY1jLZUYEGr5+njbmq8kWoP+AT2ztYt/FB7yZiPKuWeTMFfomRldCN+N
1g4OYM4qXHeDCmklpEnJRRWwhTRRzRvQKH6j3wMzoqreZjyXCS6HObMCgYEA0Q3k
8VtsKuN8HDRAQs2SMqT0c1mZ0t7MftVPS+4ziHENLy78A0gCyJuFZqhiRh/+Iqwk
zpPcXUhdZTYZSDvSpqGHubyFAB+6jwxexSxgF/OyZfbgZb0GLyviKz91xp52DE6cm
RVhTgFf455Hs0wmj3VCsJpu+VkvRe2DxCH7J5o0CgYEAnNesoRYldumfUYys1EMm
zY3E6d3s9Lzh15k30jp6NyZBVVbshn2VJ960j2wvTAElaGpe111+0naNHquDxhV6
ugvveXX/ZMqkwKc61R+tih6B+sHlBr4Y1vJDjaMl9RllPSYgKgIUN8WT2mLweFN
D00+3tpwtWTZJILP7LHoc3cCgYAUoP1df7r9xHLjyZgpv2VU0Ag7NbU0JPYi8eOg
R4IYTaojCb/+Suazav1P8v+3kcKfI00Civ6syc5FhYABk6Q0w1PEVUD+kSL0qLDJ
UjaNdb/bswX3f4102xw/i0eC1b0wfEb/gsbCnGu4ViL7dDESTRqow9/9XVu8BAXk
47YCpQKBgALLPkh1eDTTKy6qhSoG6EaQL+pR3lvu4916LehbGhvaUYftQnA9zCFV
2Gb9XlhQoJshucOL8Rql+3F4Qwa98LNYvRquZaHpw8KezYnaZKpjHeE00ZwE2wrn
RReKdczKZaLRSNg0BPhlMfQ0V6H0MnUEhmfp09dps9ShGvRlLqIi8
-----END RSA PRIVATE KEY-----
```

2 2 2 2 2 2 2 2 2 2

: wq !

After that enter the Commands : **chmod 400 file name** it is used for give the permission

```
ubuntu@ip-10-0-0-95:~$ ssh -i "vishi.pem" ubuntu@10.0.1.48
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Jul  9 04:49:07 UTC 2026

System load:  0.0           Temperature:           -273.1 C
Usage of /:   30.7% of 6.61GB Processes:                  120
```

```
Last login: Thu Jul  9
ubuntu@ip-10-0-1-48:~$
```

it change the ip address

Enter the command for update : **sudo apt update**

```
ubuntu@ip-10-0-1-48:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [272 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata [313 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64v3 Packages [291 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse Translation-en [127 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 Components [50.0 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 c-n-f Metadata [8276 B]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [316 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [83.8 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [49.3 kB]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata [5756 B]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [198 kB]
```

Enter the command to check the ping enter the command **ping google.com**

```
ubuntu@ip-10-0-1-48:~$ ping google.com
PING google.com (173.194.45.139) 56(84) bytes of data.
64 bytes from yuiadum-in-f139.1e100.net (173.194.45.139): icmp_seq=1 ttl=104 time=3.15 ms
64 bytes from yuiadum-in-f139.1e100.net (173.194.45.139): icmp_seq=2 ttl=104 time=1.90 ms
64 bytes from yuiadum-in-f139.1e100.net (173.194.45.139): icmp_seq=3 ttl=104 time=1.90 ms
64 bytes from yuiadum-in-f139.1e100.net (173.194.45.139): icmp_seq=4 ttl=104 time=1.89 ms
64 bytes from yuiadum-in-f139.1e100.net (173.194.45.139): icmp_seq=5 ttl=104 time=1.87 ms
64 bytes from yuiadum-in-f139.1e100.net (173.194.45.139): icmp_seq=6 ttl=104 time=1.87 ms
64 bytes from yuiadum-in-f139.1e100.net (173.194.45.139): icmp_seq=7 ttl=104 time=1.93 ms
64 bytes from yuiadum-in-f139.1e100.net (173.194.45.139): icmp_seq=8 ttl=104 time=1.91 ms
64 bytes from yuiadum-in-f139.1e100.net (173.194.45.139): icmp_seq=9 ttl=104 time=1.87 ms
64 bytes from yuiadum-in-f139.1e100.net (173.194.45.139): icmp_seq=10 ttl=104 time=2.97 ms
64 bytes from yuiadum-in-f139.1e100.net (173.194.45.139): icmp_seq=11 ttl=104 time=1.90 ms
```